

Upoma Saha

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Education

M.Sc. in Mechanical Engineering Grand Valley State University, Allendale, MI GPA: 3.81/4.00 Thesis: "Parametric Analysis of Injection Molding flow to Reduce Molded Parts Defects for Polycarbonate and Nylon 6/12" Thesis Supervisor: Dr. Wael Mokhtar Award: Excellence in Leadership and Service to GVSU	December 2023
B.Sc. in Materials & Metallurgical Engineering Bangladesh University of Engineering & Technology, Dhaka, Bangladesh GPA: 3.42/ 4.00 Thesis: "Making Non-Fired Bricks Using Induction Furnace Steel Slag & Electric Arc Furnace Steel Slag" Thesis Supervisor: Dr. Fahmida Gulshan	October 2018

Technical Skills & Certifications

Design:	SolidWorks, AutoCAD, GD&T
Simulation:	Star CCM+, CAMWorks, Moldex3D, ANSYS
Programming:	C, MATLAB, Python, Arduino, RSLogicx 5000
Product Design:	Prototyping, DFMEA, DFA, DFM, DOE
Certification:	CSWA – Certified SolidWorks Associate by Dassault Systeme

Teaching Experience

GRAND VALLEY STATE UNIVERSITY Visiting Faculty - Taught interdisciplinary engineering courses including Engineering Graphics, Introduction to CAD/CAM, Measurement & Data Analysis in total 12-14 credits and 100+ students per semester - Advised 30 co-op students to acquire industry knowledge per semester - Assisted the course coordinators to improve the course materials of all the assigned courses.	Grand Rapids, MI Fall 2023 – Winter 2025
Graduate Teaching Assistant - Assisted students using CNC machine, helped in soldering in the freshmen lab - Graded weekly assignments of Engineering Graphics course - Assisted students in Engineering measurement and data analysis lab and graded their weekly assignments	Fall 2021 – Winter 2023

Industry Experience

Moldex3D North America <i>Application Engineer</i> - Delivering live and virtual product demonstrations of Moldex3D software to showcase simulation capabilities in plastic injection molding processes to prospective clients. - Analyzing client-specific molding challenges and proposed tailored CAE solutions using Moldex3D, improving process optimization and defect prediction.	Farmington Hills, MI May 2025 - Present
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- Consulting simulations for customer projects in collaboration with cross-functional teams, delivering actionable insights on warpage, flow, and cooling behavior.
- Providing technical expertise during pre-sales engagements, addressing in-depth queries on simulation capabilities, material modeling, and process-specific accuracy.
- Creating and presenting customized technical proposals that align Moldex3D features with clients' injection molding process requirements
- Conducting comprehensive user training sessions post-sale to onboard clients and ensure effective use of Moldex3D
- Offering continuous post-sales technical support by troubleshooting software and process-related issues, ensuring client satisfaction and

ASPEN SURGICAL

Engineering Intern

Caledonia, MI

May 2022 - Dec 2022

- Worked in prototyping a new product project worth of \$10 million
- Assisted in designing, documenting, testing, validating following FDA compliance
- Tested manufacturability for mass production
- Organized database for CAM Works to reduce the machining process time

UNILEVER Bangladesh Limited

Production Executive

Dhaka, Bangladesh

Aug 2019 - Mar 2021

- Efficiently managed 150 contractors on-site resulting in the highest output
- Met 100% production target by leveraging leadership and technical skills.
- Increased annual production by 5% for a packaging line by troubleshooting chronic issues

Research Experience

- **Non-Fired Brick:**
Developed a non-fired brick using the industrial waste such as induction furnace slag and electric arc furnace slag. Transformed the granulated slag into powdered slag, mixed it with binding powder and cement at different percentage. A hydraulic machine was used to apply force in the mold of the brick. Instead of fire, water was used to cure the bricks for different days. The compressive strength of non-fired brick was comparatively higher than the commercial brick. **(2017-2018)**
- **Quality Improvement of Injection Molded Part:**
Developed a numerical model in Moldx3D software to analyze warpage and short-shot defects in injection molded parts made with polycarbonate and nylon 6/12 materials. Key process parameters like melt temperature, injection speed, packing pressure and coolant temperature were selected and used to analysis for Design of Experiment. The Taguchi DOE method was used to optimize those parameters. As a result, defects like warpage have been reduced by 25% and short shot by 3%. **(2022-2023)**

Publications

- Upoma Saha and Wael Mokhtar, "Quality Improvement of Polycarbonate Medical Device by Moldex3D and Taguchi DOE", *Journal of Manufacturing and Materials Processing*, Jan 2025 (<https://www.mdpi.com/2504-4494/9/1/16>)
- Upoma Saha, Afzal Hossain, and Fahmida Hossain, "Making Non-Fired Bricks Using Induction Furnace Steel Slag" in 5th IC4ME2, Bangladesh, July 2019 (<https://ieeexplore.ieee.org/author/37088341124>)

Awards

- Supply Chain Director Award 2020 – a recognition of outstanding team contribution in demonstration of superior performance as a team & in battling COVID by Unilever Bangladesh Limited in 2020
- Excellence in Leadership and Service to GVSU, fall 2023- a recognition of serving as graduate assistant for two years and volunteering in department events organizing.

Graduate Course Projects

Meclab Automation

Spring 2023

- Programmed in ladder logic, structured text & function block diagram using RS Logix 5000, controlling over ENET
- Meclab machine consists of destacker, conveyor and stacker part, has single feed and continuous feed and controlled by Allen-Bradley PLC, used ENET for networking
- All instructions were created on the HMI panel screens including the Alarms, Faults, Resets, Estop to control the parts movement
- Designed HMI screen using FactoryTalk View Machine Edition software

Aerodynamics Analysis of a Sports Car

Spring 2022

- Designed an Audi E-Tron Vision Gran Turismo in SolidWorks
- Analyzed pressure, velocity, and drag force effect on the vehicle in Star CCM+

Test Chamber Airflow Optimization

Spring 2022

- Optimization of airflow of a 14'x8' test chamber, CFD analysis was conducted Star CCM+
- The chamber inlet and outlet vent locations and opening area were changed in different combinations and the best combination was selected for test chamber to conduct experiment for crucial environment

Sumobot

Fall 2021

- Designed the robot in SolidWorks, programmed it in Arduino UNO, milled the body in a CNC machine

Practitioner Soccer Board

Fall 2021

- Made a soccer board using a vibration sensor, designed in SolidWorks, and programmed in Arduino
- Used carpentry skills to make the body and used DFMEA to improve the design. It was a simple board determining the hitting ball position with LEDs to improve soccer practitioner skills.

Memberships

- Professional Member at Society of Women Engineers (SWE) 2025
- Professional Member at American Society of Mechanical Engineers (ASME) 2025